UNIT- I

INTRODUCTION TO MACHINE LEARNING

Presenter: Ms Sakshi Madhusudan Shejole

Assistant Professor

Department of Technology SPPU

Content

- Definition of Machine Learning
- Need for Machine Learning
- Comparison between Classical Programming & Machine Learning
- AI vs ML vs DS
- Machine Learning Life Cycle
- Types of Machine Learning
- Data Preparation Techniques
- Dataset Preparation
- Key Takeaways

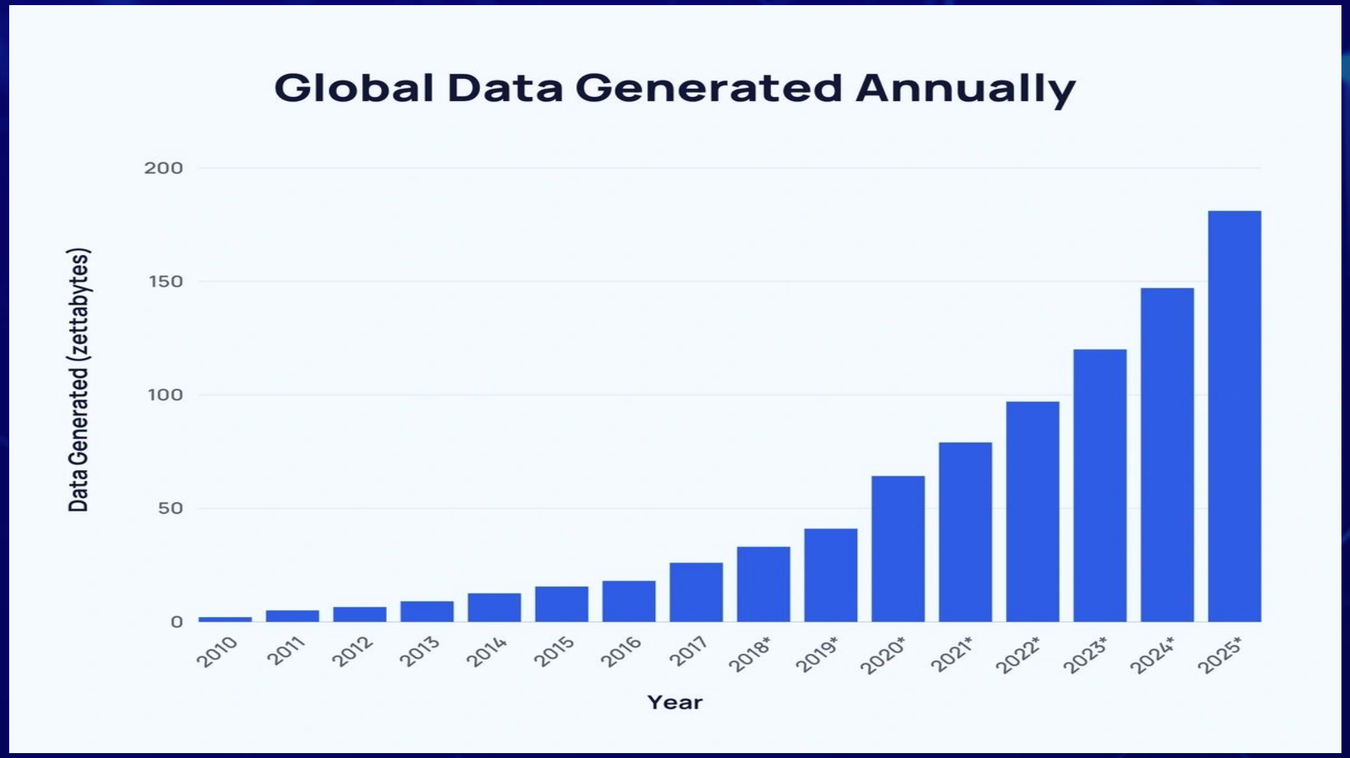
WHY ARE WE HERE?

What comes to your mind when you

hear "Machine Learning"?



Why ML is needed?



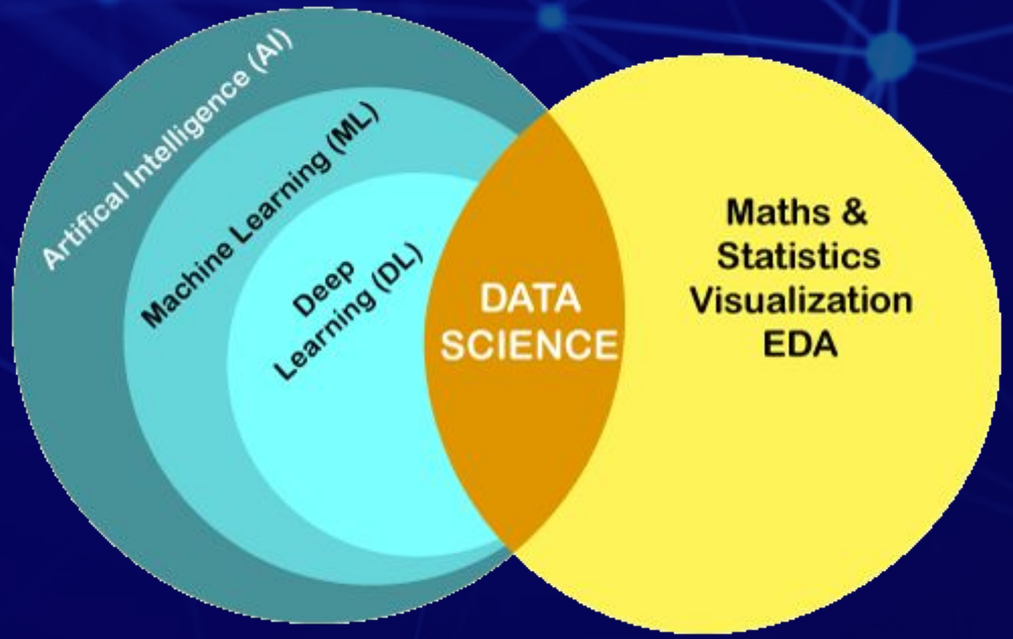
Data is growing exponentially

- Every industry generates data
- Humans cannot analyze it manually
- Machine Learning helps in D

Classical Programming vs Machine Learning

Factor	Classical Programming	Machine Learning
Instruction Method	Explicit rules and logic	Learns patterns from data
Data Handling	Structured, predictable data	Large, High-volume data, Unstructured
Outcome Predictability	Always the same result for same input	Predictions may vary as model adapts to new data
Flexibility	Limited to predefined conditions	Adjusts based on new data (self-improving)
Development Process	Linear: write, debug, deploy	Iterative: train, evaluate, tune, retrain
Transparency	Easy to trace and debug	Can be opaque, often needs explainable AI
Problem Complexity	Best for simple, well-defined tasks	Best for complex, data-rich tasks

AI vs ML vs DS



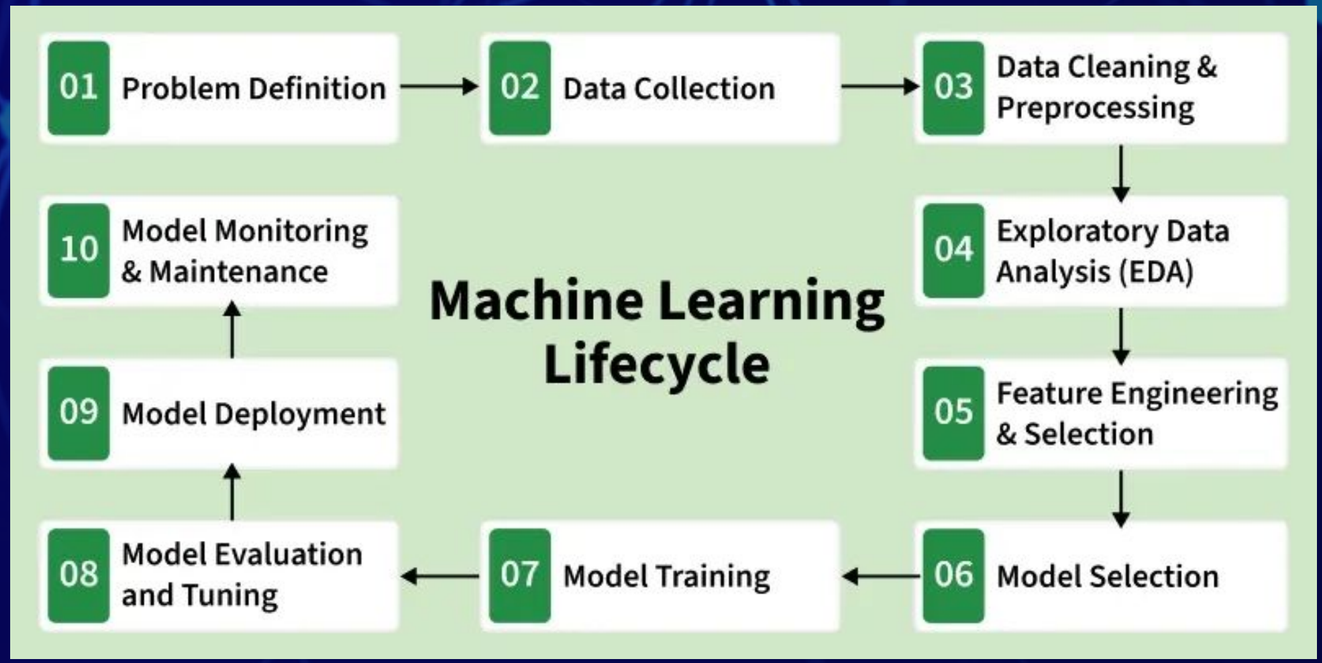
- **Artificial Intelligence (AI)**
  Making machines perform intelligent tasks.
  - **Machine Learning (ML)**
    Learning patterns from data.
  - **Data Science (DS)**
    Turning data into insights for better decisions.

What is Machine Learning?

• Machine learning is a subset of artificial intelligence where computer systems use statistical algorithms to learn patterns from data and make decisions or predictions without being explicitly programmed.

Machine Learning Life Cycle

1. Problem Definition
2. Data Collection
3. Data Cleaning & preprocessing
4. EDA
5. Feature Engineering & Selection
6. Model Selection
7. Model Training
8. Model Evaluation & Tuning
9. Model Deployment
10. Model Monitoring & Maintainance



ML Life Cycle

Example: Gmail Spam Detection

Example 1:

🎉 Congratulations! You have won ₹50,000. Click here to claim your prize.

Example 2:

Dear Students, Tomorrow's Machine Learning lecture starts at 10:00 AM.

Question: Which is Spam and Non-Spam message?

Step 1: Problem Definition

Objective: . Predict whether an incoming email is Spam or Not Spam

Available Information:

- Email Subject
- Email Content
- Sender Information
- Attachments
- Links in the Email

Expected Output:

Spam or Not Spam

Step 2: Data Collection

Data Collection phase involves systematic collection of datasets that can be used as raw data to train model. The quality and variety of data directly affect the model's performance.

Email	Correct Category
Win ₹50,000	Spam
ML Assignment	Not Spam
Free Gift Offer	Spam
Meeting Tomorrow	Not Spam

Step 3 : Data Cleaning & Pre-processing

Question: Can we directly use collected data?

If problems found then,

- **Remove duplicates**
- **Handle missing values**
- **Convert text into machine-readable format**

Step 4: Exploratory Data Analysis

Question: What patterns can we discover before training?

Observations:

Spam emails contain more hyperlinks

Unknown senders appear frequently

Words like : FREE, WINNER, URGENT

occur more often.

Step 5: Feature Engineering & Selection

Question: Can we create better information from the existing data?

Example:

- **Number of Hyperlinks**
- **Number of Capital Letters**
- **Contains "FREE"**
- **Contains "WINNER"**
- **Sender Domain**

Step 6: Model Selection

Question: Which Machine Learning algorithm should we choose?

Different Algorithms:

- **Decision Tree**
- **Logistic Regression**
- **Random Forest**
- **Naïve Bayes**

Step 7: Model Training

Question: How does the Computer learn?

Training Process:

Training Dataset

↓

Machine Learning Algorithm

↓

Learns Patterns

↓

Spam Detection Model

- Training and Testing Dataset Overview

Step 8: Model Evaluation & Tuning

Question: How do we know if our model works correctly?

If performance is poor

↓

Improve the model

↓

Tune parameters

↓

Train Again

- Overview of Accuracy, Precision, Recall, F1 Score

Step 9: Model Deployment

Question: Where will this model be used?

Deploy inside:

- **Gmail**
- **Outlook**
- **Yahoo Mail**
- **Company Email Servers**

Step 10: Model Monitoring & Maintenance

Question: Will the model always remain accurate?

It is important to

- **Collect new emails**
- **Retrain the model**
- **Improve features**
- **Update the algorithm**

- **Read complete ML Life Cycle:** https://www.geeksforgeeks.org/machine-learning/machine-learning-lifecycle/

APPLICATIONS OF ML

ASTRONOMY: DISCOVERING NEW GALAXIES

- James Webb Space Telescope generates terabytes of deep-space imagery

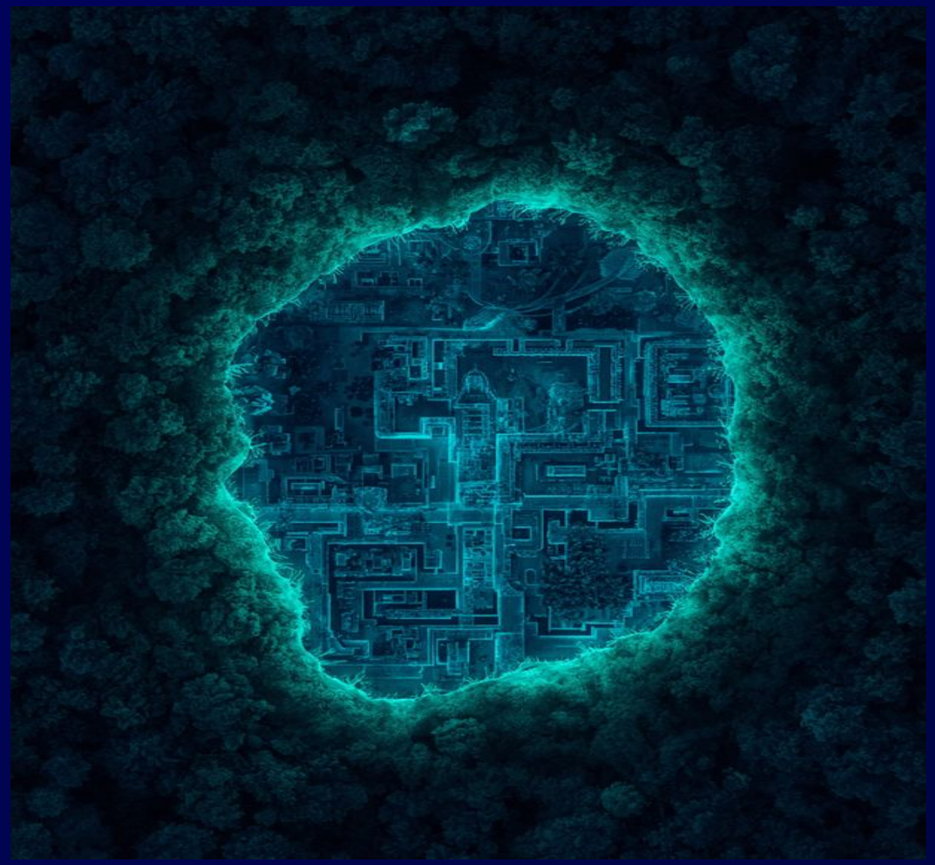
- Data Science enables discoveries no human eye could achieve alone



Alaknanda Galaxy discovered by Indian Astronomers of NCRA-TIFR

ARCHAEOLOGY: AI DISCOVERING LOST CIVILIZATIONS

- AI + LiDAR scans reveal hidden cities, temples, and ancient structures beneath dense forests without excavation.
- Satellite imagery and underwater mapping help archaeologists rediscover ancient roads, settlements, and sites like Dwarka.



AGRICULTURE: HELPING FARMERS WITH DATA

AI Crop Disease Detection:

Identify plant diseases early using image recognition and smart sensors.

Drone & Satellite Farming:

Monitor fields remotely with drones and satellite-based precision farming.

Smart Irrigation:

Optimize water usage with data-driven irrigation systems and soil sensors.

Weather& Yield Prediction:

Forecast harvests using Agri-Param models and IndiaAI Mission tools.



HEALTHCARE APPLICATIONS

Cancer Detection:

AI identifies tumors in scans with high accuracy and early-stage precision.

MRI & CT Scan Analysis:

Deep learning interprets medical images faster than human radiologists.

Disease Prediction:

Predictive models forecast patient risk using clinical and genetic data.

Drug Discovery:

ML accelerates molecule screening and personalized medicine development.



FINANCE & BUSINESS APPLICATIONS

Fraud Detection:

Detect credit card & UPI fraud in real-time using ML algorithms.

Stock Market Analytics:

Predict trends and optimize investment strategies with data models.

Recommendation Systems:

Personalize customer experiences using behavioral data insights.

Loan Approval & BI:

Automate credit scoring and drive business intelligence decisions.

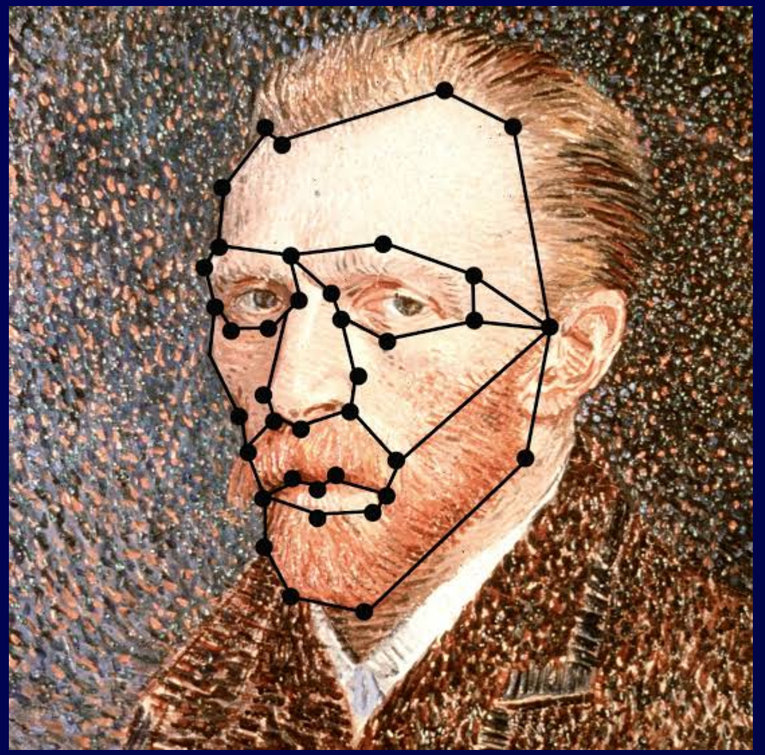


ARTS & CULTURE

Data Science helps preserve culture through artwork restoration, forgery detection, artist identification, classical dance pose detection, mudra recognition, and digital heritage preservation.



Choreography with AI: Can Machines create Dance ?



Is this painting a lost Masterpiece or fraud?

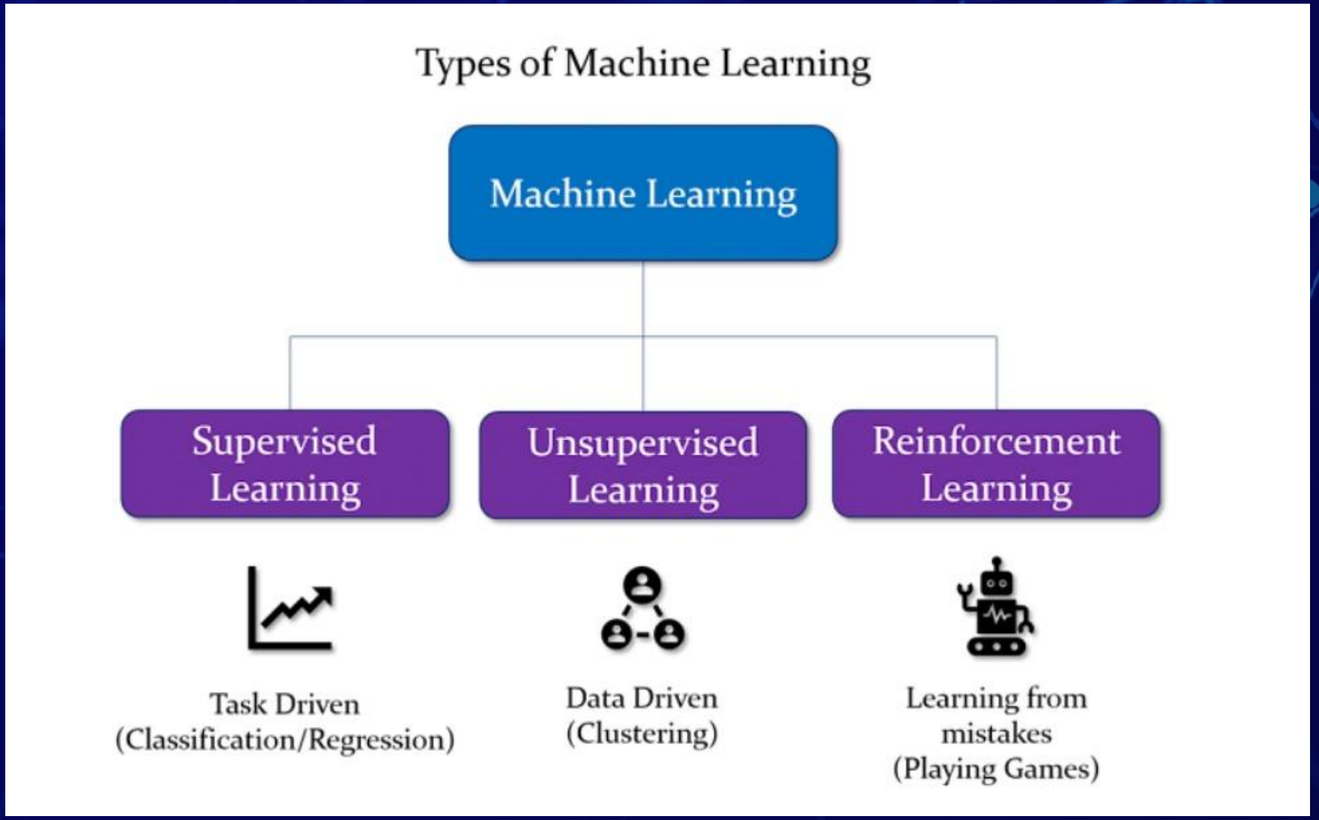
Types of Machine Learning

Question: Does every Machine Learning problem have the same type of data?

Scenarios:

- **Predict house price**
- **Group Similar Customers**
- **Train a Robot to play Football**

Types of Machine Learning



Types of ML: https://medium.com/@sandibesen/overview-of-the-types-of-machine-learning-ml-3a090b45bc4f

Supervised Learning

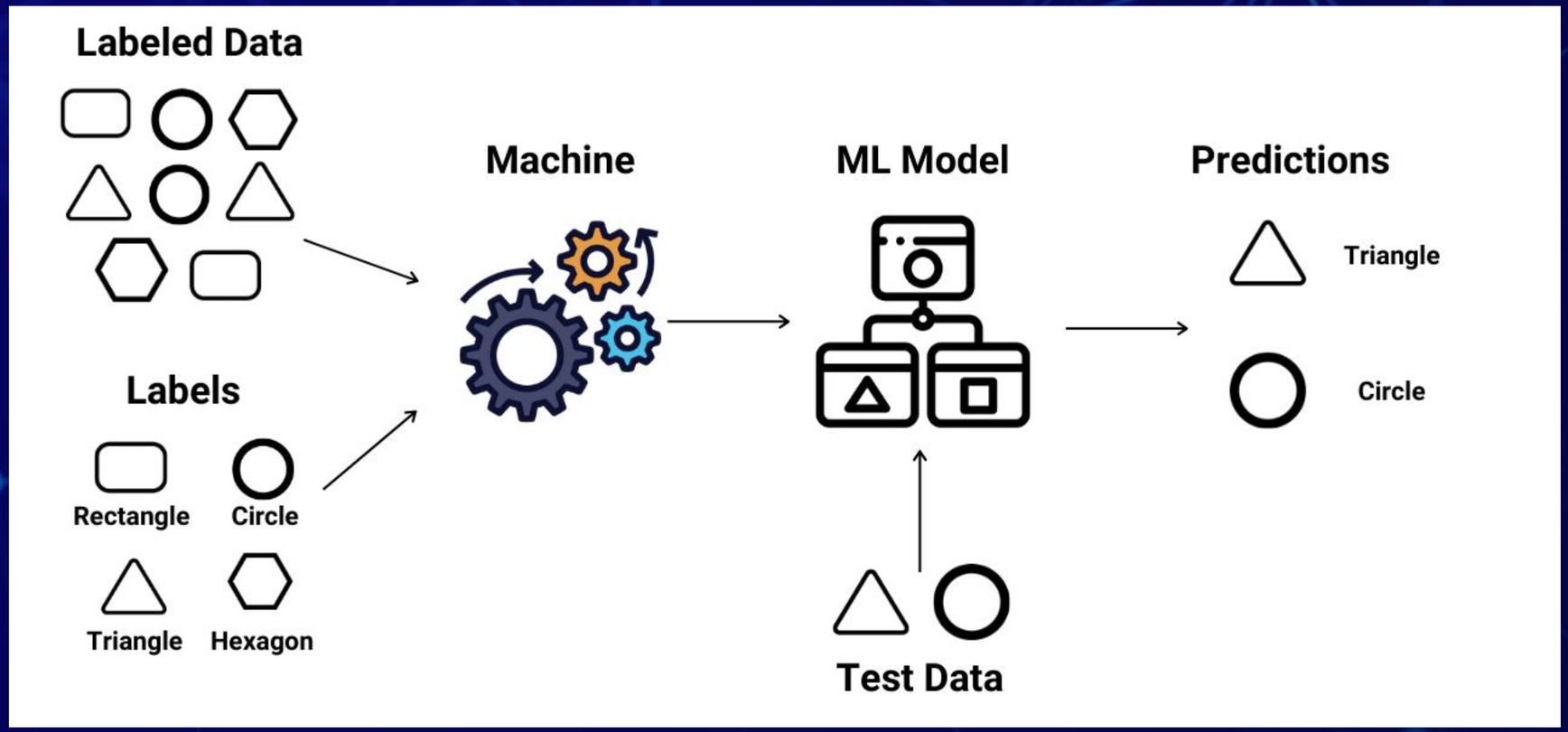
- Supervised learning is a machine learning approach where an algorithm learns from a labeled dataset to make predictions

- Supervised learning relies on datasets where both the input features and the desired output (the "label") are known.

- The algorithm learns from this labeled data to identify patterns and relationships between inputs and outputs. This process is called training.

- Once trained, the algorithm can then predict outputs for new, unseen data based on the patterns it learned.

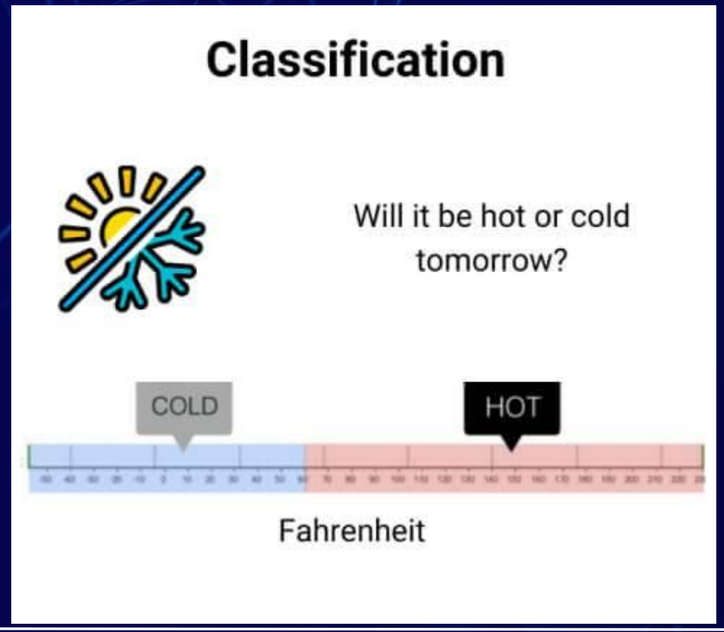
Supervised Learning Models



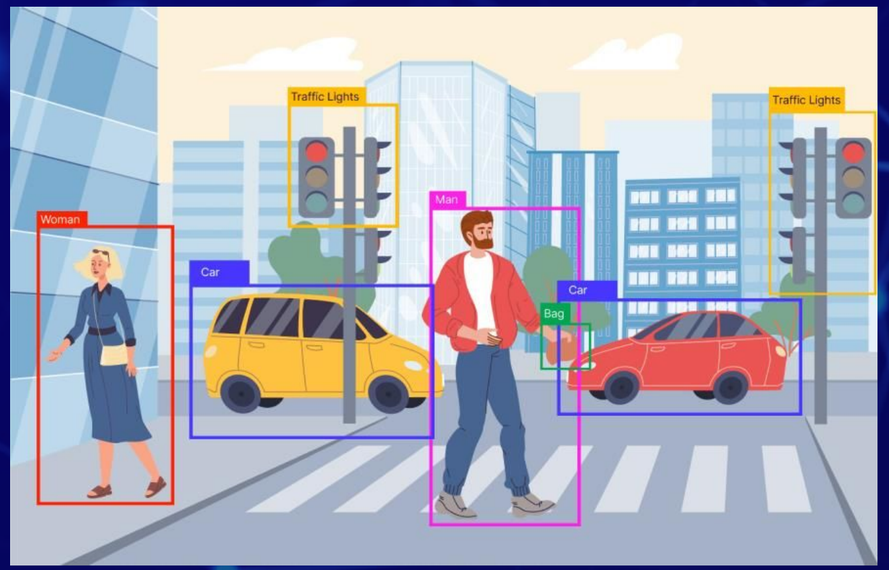
Classification

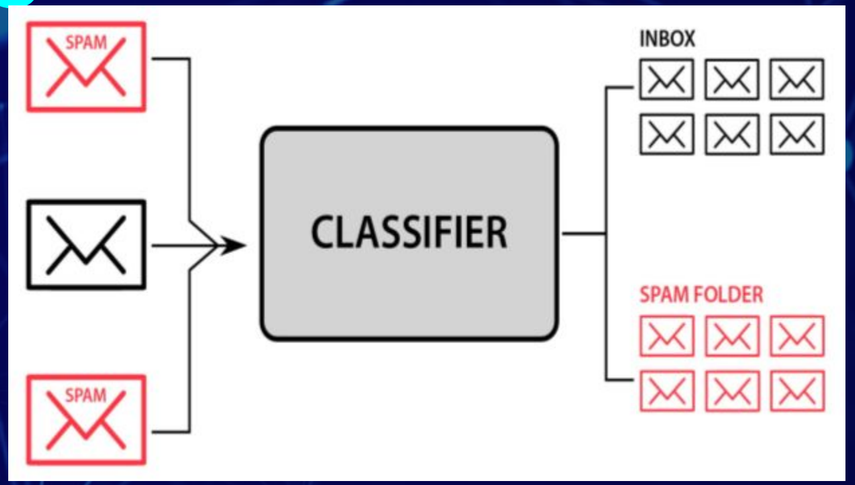
- Classification is a key task in supervised machine learning that involves predicting discrete categories or labels for data points.

- It involves training a model on labeled data to learn the relationship between input features and their corresponding categories, enabling the model to predict the class of new, unseen data.



Classification Applications





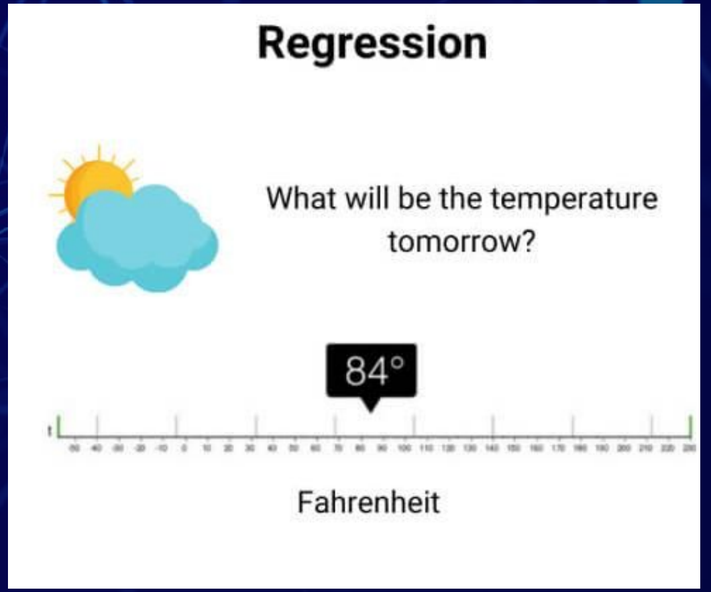


Reference: based on type of data

https://medium.com/@Sarvandani/top-applications-of-classification-in-machine-learning-e7b4351f64eb

Regression

- In statistical modeling, regression analysis is a set of statistical processe for estimating the relationships between
- a dependent variable (often called as 'label' in machine learning) and
- one or more independent variables (often called as 'features')



Regression Applications







Reference:

https://www.linkedin.com/pulse/applications-regression-analysis-various-fields-a-z-awe-ph-d-mba-xp7of/

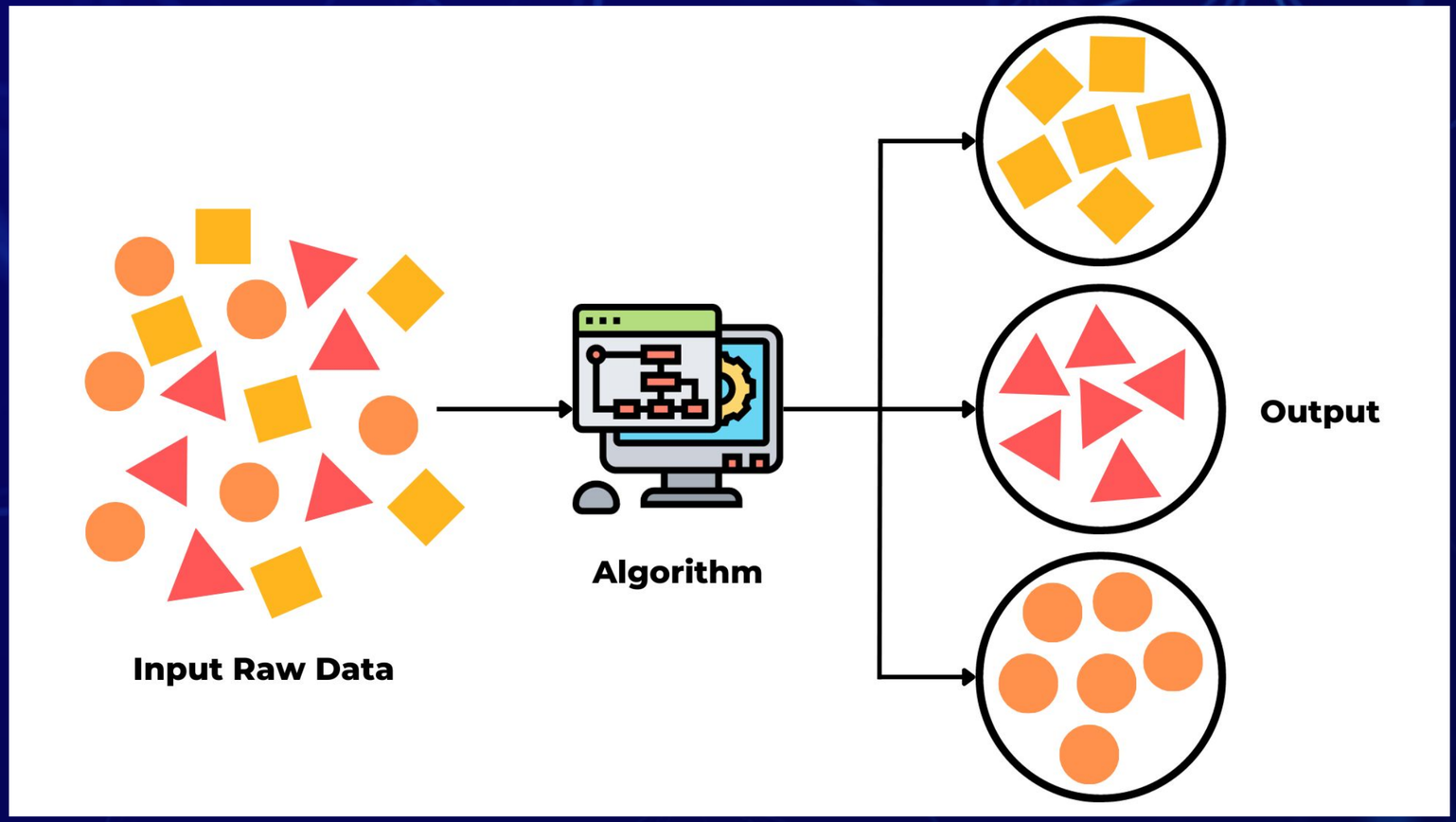
Unsupervised Learning

- Unsupervised machine learning is a type of machine learning where algorithms learn from unlabeled data without explicit guidance.

- The goal is to discover hidden patterns, structures, and relationships within the data.

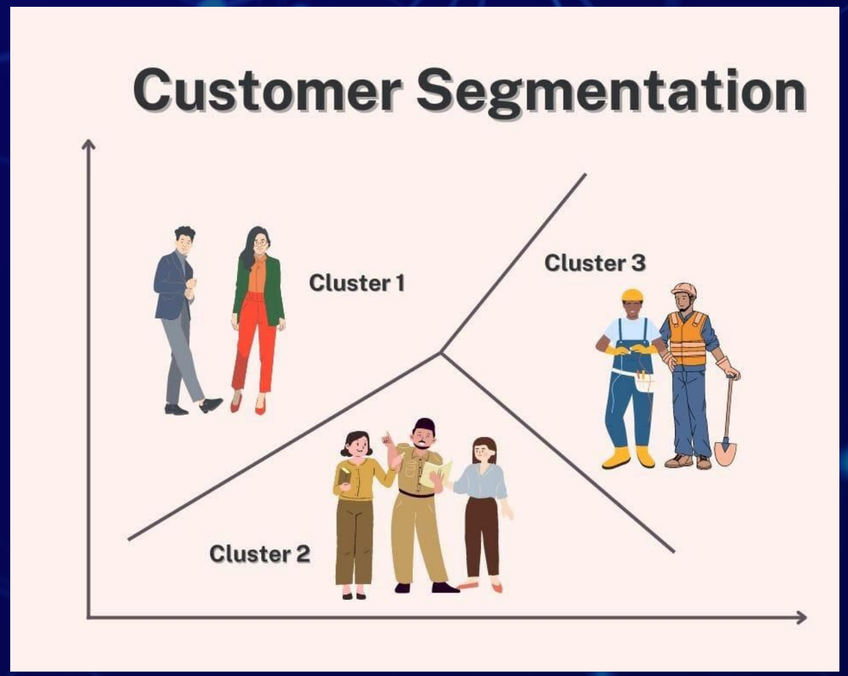
- Unlike supervised learning, the algorithm does not have a "teacher" or a set of correct answers (supervision) to learn from.

Unsupervised Learning



Unsupervised Learning Applications



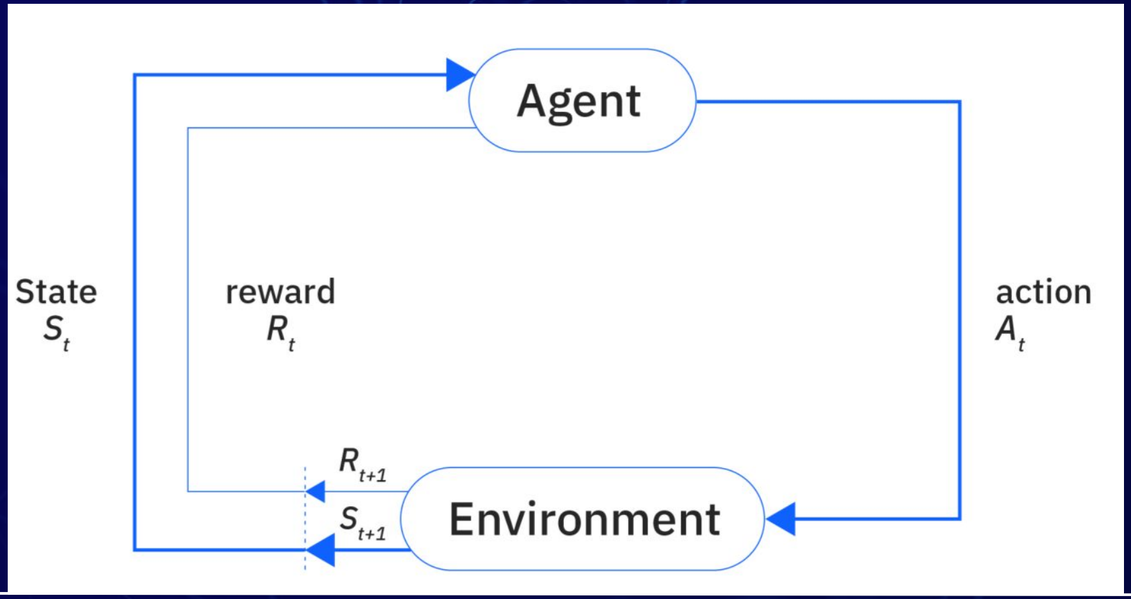


Reference: https://medium.com/@amit25173/unsupervised-learning-example-in-real-life-749cc9f4b998

Reinforcement Learning

- Reinforcement Learning is a type of machine learning where an agent learns to make decisions by interacting with an environment, taking actions, and receiving rewards or penalties.

- Unlike supervised learning, no labeled data is provided. The agent learns from experience.



Core Components

- **Agent:** The decision-making entity that interacts with the environment.
- **Environment**: The external world where the agent operates, providing states and feedback.
- **Actions**: The choices the agent can make within the environment.
- **States**: The current situation or context in which the agent finds itself.
- **Reward**: Feedback signal that indicates how good or bad an action was in a given state.

Reference: https://medium.com/@hassaanidrees7/reinforcement-learning-in-real-world-applications-from-theory-to-practice-2f67a6f673cb

Data Preparation Techniques

• Before training a Machine Learning model, the collected data must be cleaned and transformed into a suitable format. This process is known as Data Preparation.

<u>Techniques Covered:</u>

- Handling Missing Values
- Handling Outliers
- Feature Scaling
- Normalization
- Standardization
- Dataset Preparation
- Cross Validation (Introduction)

Handling Missing Values

Student	Age
Rahul	20
Sneha	?
Amit	21

Methods to Handle Missing Values:

- **Remove the record**
- **Replace with an appropriate value**
- **Collect the missing information (if possible)**

Handling Outliers

Q. What are Outliers?

Outliers are unusually high or low values that differ significantly from the rest of the data.

Q. Why handle outliers?

- May affect model performance
- Can produce inaccurate predictions

Employee	Salary
Rina	25000
Tina	27000
Mina	28000
Mita	100000

Dataset Preparation

- Before training the Machine Learning model, the dataset is divided into different parts.

Types of Datasets:

- Training Dataset: Learns the patterns from data.
- Validation Dataset: Used to improve and tune the model.
- Testing Dataset: Used to evaluate the final performance of the trained model.

Key Takeaways

- Machine Learning enables computers to learn patterns from data and make intelligent predictions.
- High-quality data preparation is essential for building accurate Machine Learning models.
- Proper dataset splitting and validation help evaluate model performance on unseen data.